Scalar Likelihood Method for Probabilistic Partial Least Squares Model with Rank n Update

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Abstract. Probabilistic Partial Least Squares (PPLS) extends Partial Least Squares (PLS) by incorporating a probabilistic framework and identifiability constraints. PPLS models the relationship between two datasets as the sum of a joint component comprising correlated latent vectors and a noise component of isotropic normal random vectors. However, estimating PPLS model parameters involves a challenging nonconvex constrained optimization problem. We propose the Scalar Likelihood Method (SLM) to estimate the parameters of the PPLS model using a scalar likelihood function of the observed variables, derived via a novel rank n update technique. This technique avoids introducing additional constraints that would compromise the model's dimension reduction properties. In addition, we derive a closed-form solution for the noise distribution in the observed data, significantly reducing parameter coupling in the objective function, thereby greatly improving parameter estimation quality. Through simulation studies, as well as association analysis and prediction tasks, we demonstrate the effectiveness and efficiency of SLM, highlighting its potential in practical applications.

1 Introduction

Partial Least Squares (PLS) [30] is a statistical method for dimensionality reduction and data analysis that combines aspects of principal component analysis (PCA) [31] and canonical correlation analysis (CCA) [16]. PLS projects input variables into a lower-dimensional latent space that maximizes the covariance between input and output variables, effectively reducing computational complexity and sample requirements. NIPALS [29] and SIMPLS [4] are commonly used to calculate the PLS parameters. PPLS extends conventional PLS by handling missing values, supporting mixture models for complex problems, and enabling Bayesian model selection to mitigate overfitting.

Estimating PPLS model parameters is challenging due to latent variables and the complex likelihood function. The EM (Expectation Maximization) algorithm [6] is the most common approach to estimating the PPLS parameters, based on the complete likelihood function. However, the EM algorithm follows a fixed alternating optimization scheme, which limits the choice of optimization strategies. In contrast, the scalar formulation allows practitioners to select different optimization algorithms based on specific computational requirements. Moreover, it is computationally expensive and sensitive to latent variable initialization. We propose the Scalar Likelihood Method (SLM) based on the scalar likelihood function of the observed variables. By focusing on observed variables, SLM reduces complexity and sensitivity to initialization while directly optimizing the observed likelihood, potentially leading to faster convergence and better model fit than EM. The main contributions of the paper are:

1. We propose a novel approach based on the scalar likelihood function of the observed variables to efficiently estimate the parameters of the PPLS model. By extending the rank one update technique to the rank n update technique, we derive the scalar log-likelihood function without introducing any additional constraints that would hinder the dimension reduction property of PPLS.
2. We derive closed-form formulas to estimate the noise distribution in the observed data. We assume that the noise follows a multivariate normal distribution, and the maximum likelihood estimate is used to obtain the scaling factor of the isotropic covariance matrix. Our method significantly reduces the parameter coupling in the objective function, and thereby greatly improves the quality of the estimated parameters.
3. We develop two applications of PPLS with an association analysis framework and a prediction method. In this framework, we demonstrate the superior performance of SLM in detecting a higher number of gene-protein pairs at various significance levels, potentially uncovering new biological insights. For the prediction method, we used the conditional distribution of a multivariate Gaussian to predict the target variables for new samples. We demonstrate the effectiveness through a simulation study with well-calibrated uncertainty estimates across varying confidence levels.

2 Related Work

PLS is a versatile method for modeling relationships between two sets of variables by projecting them into a new latent space. PLS and its variants are widely used in various research fields, including PLS regression [15] to handle multicollinearity in linear regression, PLS discriminant analysis (PLS-DA) [1] for classification tasks, sparse PLS [3] for simultaneous variable selection and dimension reduction, and kernel PLS [22] for capturing nonlinear relationships. These methods have found broad application in chemometrics, bioinformatics, and machine learning [2, 17, 12, 23, 27, 24].

The PPLS model incorporates latent variables and noise components [8, 9, 10, 18, 33], offering improved robustness and reliability compared to PLS, with recent extensions with deep learning [13]

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and omics analysis [9, 14]. The EM algorithm [6] is commonly used to estimate PPLS model parameters, alternating between computing expected values of latent variables (E-step) and maximizing the likelihood function (M-step). While robust to outliers and capable of handling missing data, EM has some limits, such as dependence on initial values and computational complexity. Several improvements have been proposed for parameter estimation for PPLS, including the SAGE [11], Monte Carlo EM [28], ECM [20], and ECME [19] algorithms, aiming to improve convergence rate, stability, and flexibility.

SAGE [11] updates parameters sequentially, Monte Carlo EM [28] uses simulations for intractable E steps, ECM [20] employs conditional maximization, and ECME [19] directly maximizes observed-data likelihood, but all remain within the EM framework. While these variants offer certain improvements over the original EM method, they remain within the EM framework and still rely on latent variable estimation. In contrast, our proposed SLM approach operates directly on the scalar likelihood function of the observed variables, providing a fundamentally different approach that offers computational advantages and improved numerical stability compared to EM-based methods.

Likelihood functions are essential for statistical inference of latent variable models, often requiring numerical optimization for an analytical solution. Various algorithms can be applied to maximize likelihood functions, such as Newton-Raphson [32], quasi-Newton methods [7], and stochastic approximation methods [21]. The choice of algorithm depends on the properties of the likelihood function, including smoothness, convexity, modality, and complexity.

3 Preliminaries

We first present some preliminaries about the PPLS model and its identifiability.

3.1 PPLS Model

The PPLS model [8] relates two sets of observed variables $x \in \mathbb{R}^p$ and $y \in \mathbb{R}^q$ to two sets of latent variables $t \in \mathbb{R}^r$ and $u \in \mathbb{R}^r$. In PPLS, it is assumed that x and y are contaminated by independent random noises $e \in \mathbb{R}^p$ and $f \in \mathbb{R}^q$, respectively, and that both t and u are affected by random noise $h \in \mathbb{R}^r$. The identity matrix is denoted by I, or I_m with m indicating the dimension. The PPLS model can be expressed as follows:

$$x = tW^\top + e,\ y = uC^\top + f,\ u = tB + h, \tag{1}$$

where x, e, y, and f are row vectors of length p, q, q, and q, respectively, and t and u are row vectors of length r; t and u are the joint parts, which are linked by the third relation of (1); e and f are mutually independent noises; W and C are the loading matrices of size $p \times r$ and $q \times r$, respectively; e and f follow isotropic normal distributions: $e \sim \mathcal{N}(0, \sigma_e^2 I_p),\ f \sim \mathcal{N}(0, \sigma_f^2 I_q)$; t follows an anisotropic distribution: $t \sim \mathcal{N}(0, \Sigma_t)$, where $\Sigma_t = \mathrm{diag}(\theta_{t_1}^2, ..., \theta_{t_r}^2)$; $B = \mathrm{diag}(b_1, ..., b_r)$ is a diagonal matrix connecting t and u; $h = [h_1, ..., h_r]$ is the noise with isotropic normal distribution: $h \sim \mathcal{N}(0, \sigma_h^2 I_r)$. With an abuse of notation, we use the same letter both for a realization and for the associated random variable. The joint distribution of (x, y) is $\mathcal{N}(0, \Sigma)$ with probability density function

$$f(x, y) = (2\pi)^{-\frac{p+q}{2}} (det\ \Sigma)^{-\frac{1}{2}} \exp\left((x, y)\Sigma^{-1}(x, y)^\top\right), \tag{2}$$

where

$$\Sigma = \begin{pmatrix} W\Sigma_t W^\top + \sigma_e^2 I_p & W\Sigma_t BC^\top \\ CB\Sigma_t W^\top & C(B^2\Sigma_t + \sigma_h^2 I_r)C^\top + \sigma_f^2 I_q \end{pmatrix}. \tag{3}$$

From observations $(x_i, y_i)_{1 \le i \le N}$ of (x, y), we desire to estimate the set of parameters $\theta = (W, C, B, \Sigma_t, \sigma_e^2, \sigma_f^2, \sigma_h^2)$. The log likelihood function of the observed data is

$$\ln L_N(\theta) = -\frac{N}{2}\left((p+q)\ln(2\pi) + \ln det\ \Sigma + \mathrm{tr}(S\Sigma^{-1})\right), \tag{4}$$

where

$$S = \frac{1}{N}\sum_{i=1}^{N}(x_i, y_i)^\top(x_i, y_i). \tag{5}$$

el Bouhaddani [8] considered the joint distribution of (x, y, t, u), which can be factorized as $p(x, y, t, u) = p(x|t)p(y|u)p(u|t)p(t)$ (with p indicating the probability law) and then designed an EM algorithm to obtain the parameters estimates. Motivated by the limits of EM (Sec. 1), we propose the SLM Algorithm. We need to maximize Eq.4, or equivalently, minimize.

$$\mathcal{L}(\theta) = \ln(det\,\Sigma) + \mathrm{tr}(S\Sigma^{-1}), \theta = (W, C, B, \Sigma_t, \sigma_e^2, \sigma_f^2, \sigma_h^2). \tag{6}$$

3.2 Identifiability

Identifiability is important for the PPLS model, which ensures that the parameter estimates are directly comparable between different studies, even when the signs of loading differ. This is essential for combining parameter estimation in meta-analyses of multiple studies. To ensure the identifiability of the PPLS model, the following constraints are required (see [8]):

1. The diagonal elements of B, b_k, are positive for $k = 1, ..., r$;
2. W and C have orthonormal columns: $W^\top W = I,\ C^\top C = I$;
3. $\sigma_{t_k}^2 b_k$, the diagonal elements of $\Sigma_t B$, for $k = 1, ..., r$, are strictly decreasing with k;
4. r is less than the minimum of p and q.

4 Expand $\mathcal{L}(\theta)$ to Scalar Level

We first introduce two lemmas that facilitate the computation of the determinant and inverse of the block matrices of Σ, and the trace of the matrix product $S\Sigma^{-1}$. We then articulate the process of expanding $\mathcal{L}(\theta)$, the non-constant term of the log-likelihood function, into scalar variables: see Theorem A below.

4.1 Two Lemmas

The computation of the determinant $det\,\Sigma$ and the inverse Σ^{-1} necessitates the computation of the block matrices in the upper left corner and the bottom right corner of Σ, both having the form of $AD_nA^T + kI_m$, where A corresponds to W and C in the model, D is a diagonal matrix. The following lemma shows how to calculate the determinant and inverse of such a matrix.

Lemma 1 (Determinant and Inverse of a Block Matrix in PPLS Model). Let $n \ge 1$ and D_n be an $n \times n$ diagonal matrix with positive diagonal entries $\sigma_1, \sigma_2, \ldots, \sigma_n$. Let $m > n$ and A be an $m \times n$ matrix with orthonormal columns (a_i): $a_i^T a_j = \delta_{ij}$, where δ_{ij} is the Kronecker delta that equals 1 if $i = j$ and 0 otherwise. Let

$k > 0$ be a positive constant. Then the following properties hold for the block matrix $AD_nA^T + kI_m$:

$$AD_nA^T + kI_m = k\prod_{i=1}^{n}\left(I_m + \frac{\sigma_i}{k}a_ia_i^T\right), \tag{7}$$

$$det(AD_nA^T + kI_m) = k^{m-n}\prod_{i=1}^{n}(k+\sigma_i), \tag{8}$$

$$(AD_nA^T + kI_m)^{-1} = \frac{1}{k}(I_m - A(I_n + kD_n^{-1})^{-1}A^T). \tag{9}$$

Given the column orthonormality, if we also impose row orthonormality on A, which will make A a square matrix, then the calculation of the determinant and inverse becomes simple:

$$\begin{aligned} AD_nA^T + kI_m &= A(D_n + kI_n)A^T, \\ det(AD_nA^T + kI_m) &= det(D_n + kI_n) = \prod_{i=1}^{n}(\sigma_i + k), \\ (AD_nA^T + kI_m)^{-1} &= A\,\mathrm{diag}\left(\frac{1}{\sigma_1+k}, \ldots, \frac{1}{\sigma_n+k}\right)A^T. \end{aligned}$$

However, while imposing row orthonormality on A can simplify computations, it also undermines the dimension-reduction aspect of the PPLS model by forcing the latent space dimension to match the observed sample dimension. Instead, we employ the rank-n update technique as an effective alternative, based on Lemma 1.

Under the constraint that A's columns are orthonormal, we can extend the Sherman-Morrison formula in rank one update [26] from a single pair of vectors to multiple pairs, and this leads to Lemma 1. With Lemma 1, we can compute $det(AD_nA^T + kI_m)$ and $(AD_nA^T + kI_m)^{-1}$ without requiring a closed-form expression, and without sacrificing the dimension-reduction property of the PPLS model or relying on iterative applications of the Sherman–Morrison formula.

For the calculation of $tr(S\Sigma^{-1})$, we need the following lemma.

Lemma 2 (Trace of Matrix Product with Diagonal Matrix). Let A,B,C be three matrices of dimension $n \times m$, $m \times r$ and $n \times r$ respectively. Let M be an $r \times r$ diagonal matrix. Then

$$\mathrm{tr}(ABMC^T) = \sum_{j=1}^{r} M_j \sum_{i=1}^{n} C_{ij} \sum_{k=1}^{m} A_{ik}B_{kj}. \tag{10}$$

It states that the trace of the product of the four matrices A, B, M, and C^T can be expressed as a triple sum of products of their entries. The proof is elementary and is therefore omitted.

4.2 Scalar Likelihood Function

Parameter estimation is challenging due to the complexity of numerically optimizing $\ln L_N(\theta)$ in the matrix form of (4). For convenience, we ignore the constant part and reverse the sign of $\ln L_N(\theta)$ of (4), and focus on the objective function defined by (6). Here we present a theorem which gives a closed form of the likelihood function, in terms of the scalar components. This theorem will facilitate the computation and optimization of the PPLS model.

To calculate the matrix product $S\Sigma^{-1}$, we partition S as follows:

$$S = \begin{pmatrix} S_{xx}(p \times p) & S_{xy}(p \times q) \\ S_{yx}(q \times p) & S_{yy}(q \times q) \end{pmatrix}. \tag{11}$$

The (j,k)-th entry of the matrix S_{xx} will be denoted by $S_{xx,jk}$, and similarly for the other matrices.

Let $D_i = (\sigma_f^2+\sigma_h^2)(\theta_{t_i}^2+\sigma_e^2)+b_i^2\theta_{t_i}^2\sigma_e^2$. Then the latent variables related to θ are the diagonal matrices $M_l = \mathrm{diag}(M_l(1), \cdots M_l(r))$, where $l = 2, 4, 6$,

$$\begin{aligned} M_2(i) &= (\sigma_f^2 + \sigma_h^2)\theta_{t_i}^2/D_i, \\ M_4(i) &= (\sigma_h^2(\theta_{t_i}^2 + \sigma_e^2) + b_i^2\theta_{t_i}^2\sigma_e^2)/D_i, \\ M_6(i) &= b_i\theta_{t_i}^2/D_i, \end{aligned}$$

and the scalars $K_l(i)$ determined by S, $\mathbf{W}$ and $\mathbf{C}$:

$$\begin{aligned} K_2(i) &= \frac{1}{\sigma_e^2}\sum_{j=1}^{p}\sum_{k=1}^{p} W_{ji}S_{xx,jk}W_{ki}, \\ K_4(i) &= \frac{1}{\sigma_f^2}\sum_{j=1}^{q}\sum_{k=1}^{q} C_{ji}S_{yy,jk}C_{ki}, \\ K_6(i) &= 2\sum_{j=1}^{p}\sum_{k=1}^{q} W_{ji}S_{xy,jk}C_{ki}. \end{aligned}$$

Theorem A $\ln(det\,\Sigma)$ and $\mathrm{tr}(S\Sigma^{-1})$ can be expressed as follows:

$$\begin{aligned} \ln(det\,\Sigma) &= (p-r)\ln\sigma_e^2 + (q-r)\ln\sigma_f^2 \\ &+ \sum_{i=1}^{r}\ln\left((\sigma_f^2+\sigma_h^2)(\theta_{t_i}^2+\sigma_e^2) + b_i^2\theta_{t_i}^2\sigma_e^2\right), \end{aligned} \tag{12}$$

$$\begin{aligned} \mathrm{tr}(S\Sigma^{-1}) &= \frac{1}{\sigma_e^2}\mathrm{tr}(S_{xx}) + \frac{1}{\sigma_f^2}\mathrm{tr}(S_{yy}) \\ &- \sum_{i=1}^{r}(K_2(i)M_2(i) + K_4(i)M_4(i) + K_6(i)M_6(i)). \end{aligned} \tag{13}$$

Let us give a sketch of the proof; the details are given in the Appendix. Applying the determinant formula for block matrices to Σ and using (8) of Lemma 1, we obtain the expression of $\ln(det(\Sigma))$. By the inverse formula for block matrices to Σ and (9), we get

$$\Sigma^{-1} = \begin{pmatrix} \frac{1}{\sigma_e^2}(I_p - WM_2W^T) & -WM_6C^T \\ -CM_6W^T & \frac{1}{\sigma_f^2}(I_q - CM_4C^T) \end{pmatrix}.$$

Then by Lemma 2, we obtain the expression of $\mathrm{tr}(S\Sigma^{-1})$.

4.3 Scalar Likelihood Method

We propose the Scalar Likelihood Method (SLM) to solve the optimization problem:

$$\min_{\theta}\mathcal{L}(\theta) \quad \text{subject to} \quad W^\top W = I_r,\ C^\top C = I_r \tag{14}$$

using Theorem A. Since solving $\nabla\mathcal{L}(\theta) = 0$ yields polynomial equations of degree $\gg 4$, no closed-form solution exists. We employ interior-point optimization with multi-start initialization to address this challenge.

Interior Point Optimization We convert the equality constraints to inequality constraints as follows:

$$g_M(\theta) := \sum_{i=1}^{r}(M_i^TM_i - 1)^2 + \sum_{i=1}^{r}(M_i^TM_i)^2 \le \epsilon_s, \tag{15}$$

where $M_i \in \mathbb{R}^{d_M}$ denotes the i-th column of the matrix $M \in \{W, C\}$ and $\epsilon_s = 10^{-2}$ represents the slack parameter. The optimization problem is formulated as $\min_{\theta \in \mathcal{C}} \mathcal{L}(\theta)$ where $\mathcal{C} = \{\theta : g_W(\theta) \leq \epsilon_s, g_C(\theta) \leq \epsilon_s\}$ defines the feasible region.

The interior-point method offers several advantages for our problem: (i) it maintains $\theta_k \in \mathrm{int}(\mathcal{C})$ for all iterations k, (ii) it exhibits superlinear convergence under the Linear Independence Constraint Qualification (LICQ), and (iii) it provides robust handling of the non-convex objective function $\mathcal{L}(\theta)$. We define the following termination criteria for the iteration sequence $\{\theta_k\}_{k=0}^{\infty}$:

$$(1)\ k \geq K_{\max} = 2 \times 10^2; \quad (2)\ \mathcal{L}(\theta_k) \leq L_{\min} = -10^{200};$$
$$(3)\ \|\nabla \mathcal{L}(\theta_k)\| \leq \tau_\nabla = 10^{-2};\ (4)\ |\mathcal{L}(\theta_k) - \mathcal{L}(\theta_{k-1})| \leq \tau_\Delta = 10^{-2}.$$

Criterion (3) ensures first-order optimality while criterion (4) detects convergence in function values. The feasibility mode handles the case where the initial point satisfies $\theta_0 \notin \mathcal{C}$. We solve the quadratic programming subproblems using the conjugate gradient method.

Multi-Start Strategy The objective function $\mathcal{L}(\theta)$ is non-convex due to bilinear terms such as $W^\top \Sigma_t W$ and multiplicative dependencies among parameters. This non-convexity creates multiple local minima, which necessitates a global optimization approach via a multi-start strategy. We generate a set of initial points $\mathcal{M} = \{\theta_0^{(1)}, \ldots, \theta_0^{(m)}\}$ where each $\theta_0^{(i)} \in \mathbb{R}^{(p+q+2)r+3}$ is constructed using the following algorithm:

Algorithm 1 Generate initial point θ_0

1: Generate $\tilde{W} \sim \mathcal{N}(0, I_p)^{p \times r}$ and $\tilde{C} \sim \mathcal{N}(0, I_q)^{q \times r}$.
2: Compute $W \leftarrow \mathrm{orth}(\tilde{W})$ and $C \leftarrow \mathrm{orth}(\tilde{C})$ via QR decomposition.
3: Initialize $\Sigma_t \leftarrow I_r$ and $B \leftarrow I_r$.
4: Set $\sigma_e^2, \sigma_f^2, \sigma_h^2 \leftarrow 0.01$.
5: Construct $\theta_0 \leftarrow [\mathrm{vec}(W), \mathrm{vec}(C), \mathrm{diag}(\Sigma_t), \mathrm{diag}(B), \sigma_e^2, \sigma_f^2, \sigma_h^2]$.
6: **Return** θ_0.

The operator $\mathrm{orth}(\cdot)$ denotes QR-based orthonormalization that ensures $W^\top W = I_r$ and $C^\top C = I_r$. This procedure generates initial points that sample uniformly from the Stiefel manifolds $\mathrm{St}(r,p) \times \mathrm{St}(r,q)$. The SLM algorithm optimizes $\mathcal{L}(\theta)$ directly without requiring latent variable marginalization, thereby achieving superior computational efficiency and numerical stability compared to EM-based methods.

5 Estimating the Noise's Distribution

We propose a pre-estimation method for the noise distribution to improve the precision and efficiency of parameter estimation in the PPLS model. By utilizing the statistical characteristics of the observed samples, the variances σ_e^2 and σ_f^2 of the noise components e and f can be estimated separately before optimizing the likelihood function $\mathcal{L}(\theta)$, and used as prior knowledge. This decouples the interaction between noise parameters and other parameters in $\theta \in \mathbb{R}^{(p+q+2)r+3}$, which appears in almost every term of $\ln det\, \Sigma$ and $\mathrm{tr}(S\Sigma^{-1})$. Fixing σ_e^2 and σ_f^2 reduces the complexity of optimization, mitigates the impact of local minima and saddle points, simplifies gradient calculation, and reduces the nonlinearity of the problem.

Let the observed data be $x_i = s_i + e_i$, where s_i is the signal and $e_i \sim \mathcal{N}(0, \sigma_e^2)$ is the noise. Given N data $\{x_i\}_{i=1}^N$, we have $\sum_{i=1}^N x_i = \sum_{i=1}^N s_i + \sum_{i=1}^N e_i$. By the law of large numbers, almost surely $\frac{1}{N}\sum_{i=1}^N e_i \to 0$, so that when N is large, $\frac{1}{N}\sum_{i=1}^N x_i \approx \frac{1}{N}\sum_{i=1}^N s_i$. Thus, $\bar{x} \approx \bar{s}$, and when $s_i \approx \bar{s}$, we have $s_i \approx \bar{x}$. The approximation $s_i \approx \bar{s}$ is reasonable under conditions such as small $\mathrm{Var}(t_i)$ or small elements of W, which reduces the difference among the s_i's. Consequently, the noise can be estimated as $e_i = x_i - s_i \approx x_i - \bar{x}$.

According to the maximum likelihood principle, the estimate $\hat{\sigma}_e^2$ of the noise variance σ_e^2 should maximize the likelihood function:

$$\hat{\sigma}_e^2 = \arg\max_{\sigma_e^2} \prod_{i=1}^N \frac{1}{(2\pi\sigma_e^2)^{d/2}} \exp\left(-\frac{1}{2\sigma_e^2}\left\|x_i - \frac{1}{N}\sum_{j=1}^N x_j\right\|^2\right) \tag{16}$$

where d is the dimension of the N observed variable x_i with N noise e_i. Notice that

$$\frac{\partial \ln L(\sigma_e^2)}{\partial \sigma_e^2} = -\frac{N}{2\sigma_e^2} + \frac{1}{2(\sigma_e^2)^2}\sum_{i=1}^N \left\|x_i - \frac{1}{N}\sum_{j=1}^N x_j\right\|^2. \tag{17}$$

Thus, the maximum likelihood estimator of the noise variance of e is:

$$\hat{\sigma}_e^2 = \frac{1}{N}\sum_{i=1}^N \left\|x_i - \frac{1}{N}\sum_{j=1}^N x_j\right\|^2. \tag{18}$$

Similarly, for N observed data $\{y_i\}_{i=1}^N$ with N noise f_i, the maximum likelihood estimator of σ_f^2 is:

$$\hat{\sigma}_f^2 = \frac{1}{N}\sum_{i=1}^N \left\|y_i - \frac{1}{N}\sum_{j=1}^N y_j\right\|^2. \tag{19}$$

Simplification of the Objective Function Since σ_e^2 and σ_f^2 are pre-estimated, the terms $(p-r)\ln\sigma_e^2$ and $(q-r)\ln\sigma_f^2$ in $\mathcal{L}(\theta)$ become constants and can be omitted from the objective function during the optimization. This simplification offers several advantages: (1) It improves computational efficiency. (2) Given that σ_e^2 and σ_f^2 are typically small, these logarithmic terms can be quite large, potentially dominating the objective function. Their removal allows for a more balanced optimization landscape, where changes in other parameters can be more evident. This can make the decrease in the objective function between successive iterations more pronounced, providing clearer indicators of optimization progress. (3) The omission of these large constant terms mitigates the risk of the function value exceeding numerical limits prematurely. While this simplification can improve the numerical stability and interpretability of the optimization, it does not alter the location of the optimum in the parameter space.

Theoretical Guarantee Under the conditions mentioned above (small $\mathrm{Var}(t_i)$ or small $\|W\|_F$), we can establish the following error bound for our noise variance estimator:

$$|\hat{\sigma}_e^2 - \sigma_e^2| \leq 2\epsilon^2 + \frac{2\sigma_e\sqrt{p}}{\sqrt{N}} \tag{20}$$

where ϵ bounds either $\sqrt{\mathrm{Var}(t_i)}$ or $\|W\|_F$. This bound shows that the estimation error decreases at a rate $O(1/\sqrt{N})$ and is well-controlled for large-scale applications, ensuring reliable initial estimates that improve the overall optimization process.

6 Time Complexity Analysis and Validation

Theoretical Analysis We analyze the time complexity of both the matrix and scalar forms of $\mathcal{L}(\theta)$ by decomposing each form into its

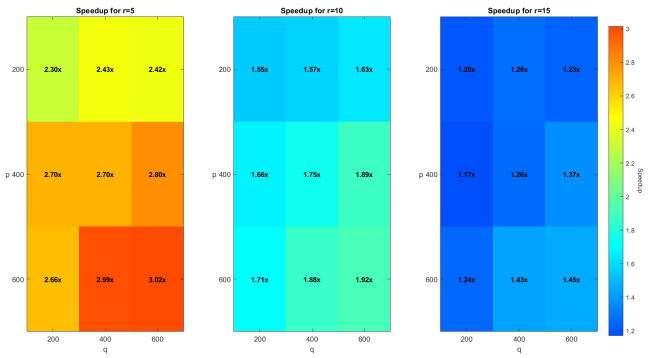

Figure 1. Heatmaps showing the speedup of the scalar likelihood function compared to the matrix likelihood function for different combinations of p, q, and r

constituent operations and computing their complexities (details provided in the Appendix). For $p, q \gg r$, the matrix form has a complexity of approximately $O(p^3 + q^3)$, while the scalar form achieves $O(rp^2 + rq^2)$. This reduction in complexity – from cubic to quadratic in p and q – highlights a significant advantage of the scalar form, especially when dealing with high-dimensional data. For example, when doubling p and q, the computation time increases by a factor of 8 with the matrix form, but only by a factor of 4 with the scalar form.

Experimental Validation To validate our theoretical analysis and demonstrate the computational efficiency of SLM, we conduct a comprehensive speed comparison between the scalar and matrix likelihood functions in various dimensions of p, q, and r. We visualize the results in Figure 1, which demonstrates the superior performance of the scalar likelihood function. The heatmaps represent the speedup factor, calculated as the ratio of the execution time of the matrix likelihood function to that of the scalar likelihood function. The key observations from the experimental results include:

1. **Consistent Speedup:** The scalar likelihood function consistently outperforms the matrix likelihood function across all tested configurations, with speedup factors ranging from 1.17 to 3.02.
2. **Dimension Dependence:** The speedup becomes more pronounced as p and q increase.

Discussion The experimental results align qualitatively with our theoretical complexity analysis, confirming the consistent advantage of the scalar form over the matrix form. The observed increase in speedup with higher p and q and lower r matches our theoretical analysis. While the quantitative speedup factors (1.17 to 3.02) may appear modest relative to the asymptotic complexity difference, several factors may explain this discrepancy:

- ignored factors, which appear in the original Big O notation;
- highly optimized matrix libraries, which may reduce the practical impact of the theoretical complexity difference;
- the range of tested dimensions, which may not fully reveal the asymptotic behavior.

Despite these considerations, our results underscore the computational advantage of SLM, especially for PPLS problems with high-dimensional data (high p and q) and small latent space (low r). The improved efficiency can lead to significant time savings in parameter estimation.

Space Complexity Analysis Beyond computational time, space efficiency is crucial for large-scale problems. The matrix form requires storing Σ, S, and Σ^{-1}, totaling $O((p+q)^2)$ space. In contrast, the scalar form stores loading matrices ($pr + qr$ elements), diagonal matrices ($3r$ elements), sample covariances ($p^2 + q^2 + pq$ elements), and intermediate variables ($3r$ elements), totaling $O(p^2 + q^2)$ space.

For $p = q = 10,000$ and $r = 10$ (double precision), the matrix form requires approximately 8.94 GB while the scalar form requires only 2.24 GB—a 75% reduction that enables processing substantially larger datasets on standard hardware.

7 Experiments and Applications

The PPLS model, once fitted to existing data, provides a powerful framework for various applications. We first use a simulation experiment to compare the EM, ECM, and SLM algorithms. We then demonstrate the versatility and effectiveness of PPLS through two distinct applications: association analysis and prediction. All the SLM, EM, and ECM algorithms are implemented in Python. The EM and ECM algorithms follow the formulation in el Bouhaddani et al. [8], utilizing closed-form solutions in the M-step. The complete source code, implementation details, and datasets used in this paper are available at https://github.com/codinghhr1103/ECAI-PPLS-SLM.git.

7.1 Simulation Experiment

We conduct a comprehensive Monte Carlo study to compare SLM, EM, and ECM algorithms on parameter recovery, convergence behavior, and statistical properties under the PPLS framework.

Experimental Design We generate $M = 100$ independent realizations where each realization follows a normal distribution: $(X^{(k)}, Y^{(k)}) \sim \mathcal{N}(0, \Sigma(\Theta_{\text{true}}))$, with dimensions $p = q = 20$, latent dimension $r = 3$, and sample size $N = 500$. The true parameter set is fixed as $\Theta_{\text{true}} = (W_{\text{true}}, C_{\text{true}}, B_{\text{true}}, \Sigma_{t,\text{true}}, \sigma^2_{e,\text{true}}, \sigma^2_{f,\text{true}}, \sigma^2_{h,\text{true}})$, where the loading matrices satisfy the orthonormality constraints $W_{\text{true}}^\top W_{\text{true}} = I_r$ and $C_{\text{true}}^\top C_{\text{true}} = I_r$ via Gram-Schmidt orthonormalization of sinusoidal patterns. For each trial k, we independently sample the stochastic components according to:

$$T^{(k)} \sim \mathcal{N}(0, \Sigma_{t,\text{true}}), \quad H^{(k)} \sim \mathcal{N}(0, \sigma^2_{h,\text{true}} I_r),$$
$$E^{(k)} \sim \mathcal{N}(0, \sigma^2_{e,\text{true}} I_p), \quad F^{(k)} \sim \mathcal{N}(0, \sigma^2_{f,\text{true}} I_q). \tag{21}$$

The observed data is generated according to the PPLS structure: $X^{(k)} = T^{(k)} W_{\text{true}}^\top + E^{(k)}$, $Y^{(k)} = (T^{(k)} B_{\text{true}} + H^{(k)}) C_{\text{true}}^\top + F^{(k)}$.

Algorithm Configuration The SLM algorithm employs the multi-start strategy (Sec. 4.3). We apply interior-point optimization to the scalar likelihood function $\mathcal{L}(\theta)$ with pre-estimated noise variances $\hat{\sigma}^2_e$ and $\hat{\sigma}^2_f$. The convergence criteria are set as maximum iterations $I^{\text{SLM}}_{\max} = 100$ and gradient tolerance $\|\nabla \mathcal{L}(\theta_k)\| \leq 10^{-2}$. The EM algorithm alternates between the E-step that computes $\mathbb{E}[\mathbf{t}_i, \mathbf{u}_i | \mathbf{x}_i, \mathbf{y}_i, \theta^{(k)}]$ and the M-step that updates $\theta^{(k+1)} \leftarrow \arg\max_\theta Q(\theta | \theta^{(k)})$. The ECM algorithm decomposes the M-step into sequential conditional maximization steps: first updating (W, C), then (B, Σ_t), and finally $(\sigma^2_e, \sigma^2_f, \sigma^2_h)$, with E-step computations performed after each update. Both EM and ECM algorithms use maximum iterations $I^{\text{EM}}_{\max} = 1000$ and relative tolerance $\frac{\|\theta_k - \theta_{k-1}\|}{\|\theta_{k-1}\| + 10^{-10}} \leq 10^{-4}$ for convergence. All algorithms employ $S = 32$ random initializations per trial to address the non-convex

optimization landscape. This choice balances computational exploration of the parameter space against computational cost, based on convergence saturation analysis that shows diminishing returns beyond this number.

Table 1. Convergence statistics across $M = 100$ Monte Carlo trials

Method	$\mathbb{E}[I]$	$\sqrt{\text{Var}[I]}$	$\min I$	$\max I$	Median$[I]$
SLM	36.2	26.3	9	100	26.5
EM	427.5	230.4	123	1000	364
ECM	442.8	242.5	102	1000	415

Convergence Analysis The convergence statistics are presented in Table 1, where I denotes the number of iterations required for convergence. The SLM algorithm achieves approximately 12-fold speedup compared to EM variants by directly optimizing $\mathcal{L}(\theta)$, thereby avoiding the computational overhead of latent-variable marginalization inherent in EM methods. The coefficient of variation is $\text{CV}_{\text{SLM}} = 0.73$ versus $\text{CV}_{\text{EM}} = 0.54$, indicating that while SLM exhibits more variable convergence paths, it consistently demonstrates superior efficiency.

Table 2. Parameter estimation accuracy: MSE $\times 10^2$ (mean $\pm$ standard deviation)

Method	MSE_W	MSE_C	MSE_B	MSE_{Σ_t}	$\text{MSE}_{\sigma_h^2}$
SLM	7.44 $\pm$1.46	7.45 $\pm$1.32	38.91 $\pm$20.31	46.32 $\pm$2.37	0.095 $\pm$0.048
EM	6.59 $\pm$2.68	4.53 $\pm$3.58	88.55 $\pm$56.67	15.60 $\pm$10.13	90.25 $\pm$0.00
ECM	5.53 $\pm$3.19	5.96 $\pm$2.74	67.80 $\pm$57.82	11.97 $\pm$10.11	90.25 $\pm$0.00

Parameter Recovery Analysis The mean squared error is reported in Table 2 with $\text{MSE}_\theta = \mathbb{E}[(\hat{\theta} - \theta_{\text{true}})^2]$ scaled by a factor of 10^2 for readability. For matrix parameters, we compute MSE as $\|\hat{M} - M_{\text{true}}\|_F^2/\dim(M)$, where $\|\cdot\|_F$ denotes the Frobenius norm. For diagonal matrices, we use $\|\text{diag}(\hat{D}) - \text{diag}(D_{\text{true}})\|_2^2/r$, where $\|\cdot\|_2$ is the Euclidean norm. The SLM algorithm demonstrates particularly superior performance in estimating σ_h^2, where EM and ECM algorithms exhibit catastrophic failure with systematic convergence to the degenerate solution $\hat{\sigma}_h^2 \to 0$. This degeneracy creates a likelihood ridge where $u = tB$, which traps the EM iterations. The SLM's explicit treatment of the covariance determinant structure $\det(\Sigma)$ maintains numerical stability and avoids such singularities.

W Recovery Analysis Figure 2 demonstrates that all three algorithms successfully recover the sinusoidal structure of the loading matrix W, validating the identifiability of the PPLS model under the orthonormality constraint $W^\top W = I_r$. The constraint restricts solutions to the Stiefel manifold $\text{St}(r, p)$, ensuring structural preservation across all methods. All algorithms achieve comparable MSE values for W estimation (ranging from 5.53 to 7.44×10^{-2}), with minor variations reflecting different optimization paths on the manifold. In SLM, the loading matrix W appears in multiple terms of the scalar likelihood: $K_2(i)$ involving $W^\top S_{xx} W$ and $K_6(i)$ capturing the cross-term $W^\top S_{xy} C$, providing direct gradient information for optimization. Despite algorithmic differences, the successful recovery of the sinusoidal patterns confirms that loading matrix estimation is well-posed under the PPLS framework with appropriate orthonormality regularization.

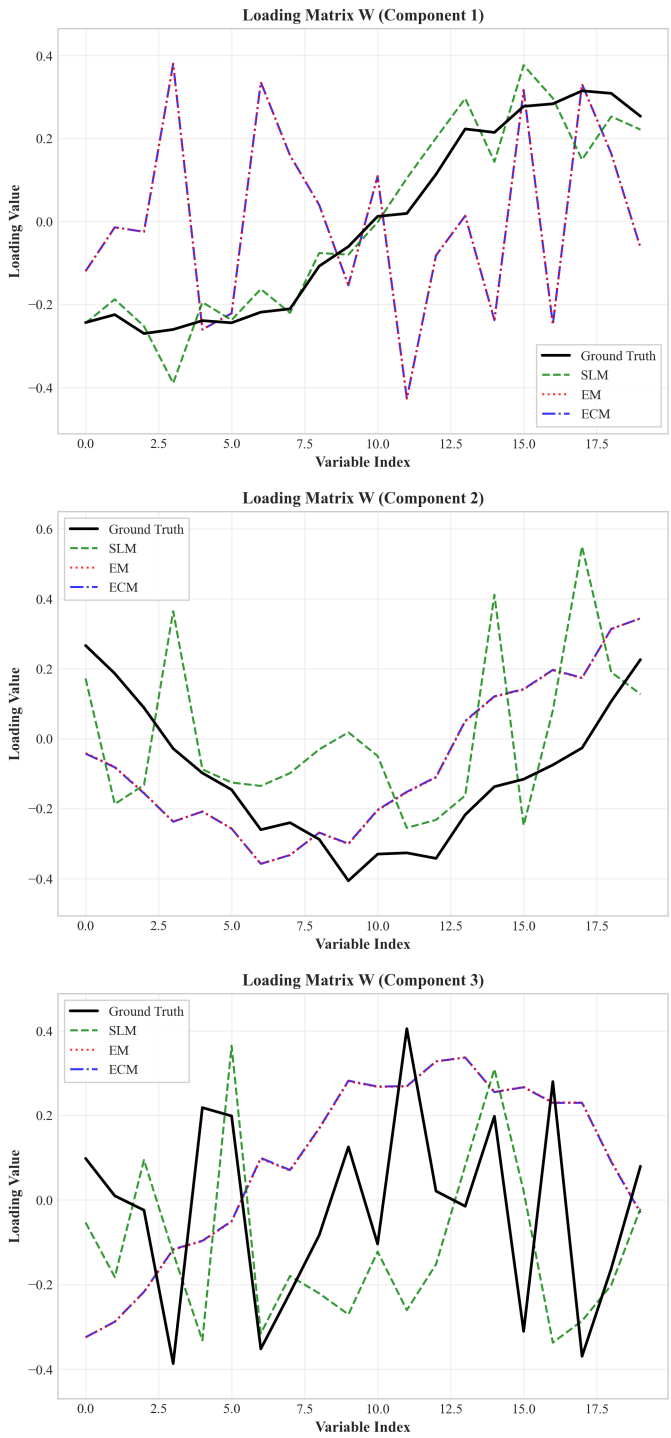

Figure 2. Recovery of loading matrix W components: Ground truth W_{true} (black), SLM estimate $\hat{W}_{\text{SLM}}$ (green), EM estimate $\hat{W}_{\text{EM}}$ (red), ECM estimate $\hat{W}_{\text{ECM}}$ (blue)

7.2 *Application in Association Analysis*

The PPLS method is particularly suited for association analysis due to its ability to handle high-dimensional data and capture latent structures that explain correlated variations across multiple datasets. We propose a novel association analysis framework using the PPLS

model, which extends beyond traditional approaches by integrating the SLM and EM algorithms to identify significant gene-protein pairs in multi-omics data.

Algorithm 2 Compute correlations and screen significant gene-protein pairs with our SLM algorithm

Require: $\mathbf{X}_s \in \mathbb{R}^{N_s \times n_{G_s}}$, $\mathbf{Y}_s \in \mathbb{R}^{N_s \times n_{P_s}}$, $\mathbf{W} \in \mathbb{R}^{n_{G_s} \times r}$, $\mathbf{C} \in \mathbb{R}^{n_{P_s} \times r}$, significance level α
Ensure: Correlation matrices $\mathbf{R}_{\mathbf{x},\mathbf{t}}$, $\mathbf{R}_{\mathbf{y},\mathbf{u}}$, P-value matrices $\mathbf{P}_{\mathbf{x},\mathbf{t}}$, $\mathbf{P}_{\mathbf{y},\mathbf{u}}$, and sets of significant gene-protein pairs

$\mathbf{T} = \mathbf{X}_s\mathbf{W}$, $\mathbf{U} = \mathbf{Y}_s\mathbf{C}$
Initialize $\mathbf{R}_{\mathbf{x},\mathbf{t}}, \mathbf{R}_{\mathbf{y},\mathbf{u}} \in \mathbb{R}^{n_{G_s} \times r}$, $\mathbf{P}_{\mathbf{x},\mathbf{t}}, \mathbf{P}_{\mathbf{y},\mathbf{u}} \in \mathbb{R}^{n_{P_s} \times r}$
for $j = 1$ to r **do**
  for $i = 1$ to n_{G_s} **do**
    Compute $\mathbf{R}_{\mathbf{x},\mathbf{t}}(i,j) = \rho(\mathbf{X}_s(:,i), \mathbf{T}(:,j))$
    Compute $\mathbf{P}_{\mathbf{x},\mathbf{t}}(i,j) = p_value(\mathbf{X}_s(:,i), \mathbf{T}(:,j))$
  end for
  for $i = 1$ to n_{P_s} **do**
    Compute $\mathbf{R}_{\mathbf{y},\mathbf{u}}(i,j) = \rho(\mathbf{Y}_s(:,i), \mathbf{U}(:,j))$
    Compute $\mathbf{P}_{\mathbf{y},\mathbf{u}}(i,j) = p_value(\mathbf{Y}_s(:,i), \mathbf{U}(:,j))$
  end for
  Initialize $\mathcal{S}_{\mathbf{x},j} = \emptyset$, $\mathcal{S}_{\mathbf{y},j} = \emptyset$
  for $i = 1$ to n_{G_s} **do**
    if $\mathbf{P}_{\mathbf{x},\mathbf{t}}(i,j) < \alpha$ **then**
      Add i-th gene to $\mathcal{S}_{\mathbf{x},j}$
    end if
  end for
  for $i = 1$ to n_{P_s} **do**
    if $\mathbf{P}_{\mathbf{y},\mathbf{u}}(i,j) < \alpha$ **then**
      Add i-th protein to $\mathcal{S}_{\mathbf{y},j}$
    end if
  end for
  Output j, $\mathcal{S}_{\mathbf{x},j}$, and $\mathcal{S}_{\mathbf{y},j}$
end for
return $\mathbf{R}_{\mathbf{x},\mathbf{t}}, \mathbf{R}_{\mathbf{y},\mathbf{u}}, \mathbf{P}_{\mathbf{x},\mathbf{t}}, \mathbf{P}_{\mathbf{y},\mathbf{u}}$

Dataset The BRCA multi-omics TCGA dataset [5] includes four types of omics data from 705 breast cancer patients. We focus on the association analysis of gene expression ($\mathbf{X} \in \mathbb{R}^{705\times604}$) and protein expression ($\mathbf{Y} \in \mathbb{R}^{705\times223}$) data due to their potential to reveal latent variable structures that could elucidate complex relationships between gene and protein expression in breast cancer.

Analysis Procedure Let $\mathbf{X} \in \mathbb{R}^{N\times n_G}$ and $\mathbf{Y} \in \mathbb{R}^{N\times n_P}$ be the gene expression and protein expression data matrices, respectively, where N is the total number of data, n_G is the total number of genes, and n_P is the total number of proteins. After preprocessing the data by removing missing values and standardizing variables, we randomly select N_s samples and n_{G_s} columns from $\mathbf{X}$ and n_{P_s} columns from $\mathbf{Y}$ to obtain two smaller matrices $\mathbf{X}_s \in \mathbb{R}^{N_s\times n_{G_s}}$ and $\mathbf{Y}_s \in \mathbb{R}^{N_s\times n_{P_s}}$. In this study, we set $N_s = 100$, $n_{G_s} = 100$, $n_{P_s} = 100$, and use $r = 10$ for latent variables. We then run the SLM and EM algorithms on these matrices using r latent variables, obtaining loading matrices $\mathbf{W} \in \mathbb{R}^{n_{G_s}\times r}$ and $\mathbf{C} \in \mathbb{R}^{n_{P_s}\times r}$, which represent the weights of the latent variables. The score matrices of the observed and target variables in the latent space are computed as:

$$\mathbf{T} = \mathbf{X}_s\mathbf{W} \in \mathbb{R}^{N_s\times r},\ \mathbf{U} = \mathbf{Y}_s\mathbf{C} \in \mathbb{R}^{N_s\times r}, \tag{22}$$

where each column of $\mathbf{T}$ and $\mathbf{U}$ represents a latent variable, and the correspondence between them is expressed by the PPLS model equation $\mathbf{U} = \mathbf{TB} + \mathbf{H}$, with $\mathbf{B}$ being the regression coefficient matrix and $\mathbf{H}$ the noise term. Algorithm. 2 computes the correlation coefficients $\rho(\mathbf{x}_i, \mathbf{t}_j)$ and their P-values $p_value(\mathbf{x}_i, \mathbf{t}_j)$ between genes, proteins, and latent variables based on the loading matrices. It could be screened out the genes and proteins significantly correlated with each latent variable based on a given significance level threshold α, forming corresponding sets $\mathcal{S}_{\mathbf{x},j}$ and $\mathcal{S}_{\mathbf{y},j}$, $j = 1, 2, ..., r$.

Table 3. Top 10 gene-protein pairs identified by SLM among all latent variables sorted by the sum of absolute correlation coefficients

LV	Gene	ρ(G, LV)	Protein	ρ(P, LV)	$\sum\|\rho\|$
9	ASCL1	0.65603	LKB1	0.58892	1.2450
6	RERGL	0.72045	Snail	-0.50774	1.2282
6	LOC28 3392	0.70708	Snail	-0.50774	1.2148
1	GRIA4	-0.69669	Rb	0.50966	1.2063
1	CRYAB	-0.68359	Rb	0.50966	1.1932
6	CIDEC	0.6827	Snail	-0.50774	1.1904
1	KRT17	-0.67543	Rb	0.50966	1.1851
6	SOX11	-0.67135	Snail	-0.50774	1.1791
1	CA4	-0.66204	Rb	0.50966	1.1717
8	COMP	0.59451	p62 LCK ligand	-0.57515	1.1697

Table 4. Number of detected gene-protein pairs by SLM and EM under different p-values

Method	$p < 1e^{-7}$	$p < 1e^{-6}$	$p < 1e^{-5}$	$p < 1e^{-4}$
SLM	63	126	227	431
EM	32	65	87	174
Overlap	0	0	4	15

In Table 3, we display the top 10 gene-protein pairs identified by the SLM algorithm with $p < e^{-7}$ for all latent variables sorted by the sum of the absolute values of the correlation coefficients, where LV stands for latent variable, $\rho(G,\ LV)$ represents the correlation coefficient between the gene and the latent variable, $\rho(P,\ LV)$ represents the correlation coefficient between the protein and the latent variable, and $\sum|\rho| = |\rho(G,\ LV)| + |\rho(P,\ LV)|$. The identified gene-protein pairs show diverse correlation strengths, including both positive and negative associations. The results suggest complex regulatory relationships in biological processes. This new perspective warrants further experimental validation to understand the roles of these pairs in disease development, potentially informing new therapeutic strategies.

In Table 4, we compare the SLM algorithm with the EM algorithm in the above-mentioned application. The results demonstrate that our SLM algorithm consistently detects a higher number of gene-protein pairs across all significance levels than the EM algorithm. Interestingly, there is minimal overlap between the pairs detected by the two methods, especially at higher significance levels. This suggests that the SLM and EM algorithms may be capturing different aspects of the underlying biological relationships. Our SLM algorithm shows higher sensitivity in detecting potential gene-protein associations. The greater number of detected pairs by the SLM algorithm could discover novel biological insights that might be missed by the EM algorithm.

7.3 Application in Prediction

We propose a prediction method based on the estimated PPLS model, showcasing how PPLS can be used to predict target variables corresponding to the new samples of the independent variables. Let $\mathbf{x}_{new} \in \mathbb{R}^{1\times p}$ be the new observation data and $\mathbf{y}_{new} \in \mathbb{R}^{1\times q}$ be the corresponding unknown target data. We can predict $\mathbf{y}_{new}$ by computing the mean of the conditional distribution of $\mathbf{y}_{new}$ given $\mathbf{x}_{new}$ and the estimated model parameters $\hat{\mathbf{\Theta}}$:

$$\begin{aligned}\hat{\mathbf{y}}_{new} &= \mathbb{E}[\mathbf{y}_{new}|\mathbf{x}_{new}, \hat{\mathbf{\Theta}}] \\ &= \int \mathbf{y}_{new} \cdot p(\mathbf{y}_{new}|\mathbf{x}_{new}, \hat{\mathbf{\Theta}}) d\mathbf{y}_{new}. \end{aligned} \tag{23}$$

To compute the conditional distribution $p(\mathbf{y}_{new}|\mathbf{x}_{new}, \hat{\mathbf{\Theta}})$, we use the following property of multivariate Gaussian distributions[25]:

Property 1 (Conditional Distribution of Multivariate Gaussian). For a multivariate Gaussian variable

$$\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix}\mu_x\\ \mu_y\end{pmatrix}, \begin{pmatrix}\mathbf{\Sigma}_{xx} & \mathbf{\Sigma}_{xy}\\ \mathbf{\Sigma}_{yx} & \mathbf{\Sigma}_{yy}\end{pmatrix}\right), \tag{24}$$

the conditional distribution of the column vector $\mathbf{y}$ given the column vector $\mathbf{x}$ is also a multivariate Gaussian distribution:

$$\mathbb{E}[\mathbf{y}|\mathbf{x}] = \mu_y + \mathbf{\Sigma}_{yx}\mathbf{\Sigma}_{xx}^{-1}(\mathbf{x} - \mu_x), \tag{25}$$

$$\mathrm{Cov}[\mathbf{y}|\mathbf{x}] = \mathbf{\Sigma}_{yy} - \mathbf{\Sigma}_{yx}\mathbf{\Sigma}_{xx}^{-1}\mathbf{\Sigma}_{xy}. \tag{26}$$

From the PPLS model, the joint distribution of the observation data $\mathbf{X}$ and target data $\mathbf{Y}$ is $\mathcal{N}(0, \Sigma)$ with Σ defined in (3). Applying Property 1 to the PPLS model, the mean and covariance matrix of the conditional distribution of $\mathbf{y}_{new}$ given $\mathbf{x}_{new}$ are:

$$\mathbb{E}[\mathbf{y}_{new}|\mathbf{x}_{new}, \hat{\mathbf{\Theta}}] = \hat{\mathbf{C}}\hat{\mathbf{B}}\hat{\mathbf{\Sigma}}_t\hat{\mathbf{W}}^T(\hat{\mathbf{W}}\hat{\mathbf{\Sigma}}_t\hat{\mathbf{W}}^T + \hat{\sigma}_e^2\mathbf{I})^{-1}\mathbf{x}_{new}, \tag{27}$$

$$\begin{aligned}\mathrm{Cov}[\mathbf{y}_{new}|\mathbf{x}_{new}, \hat{\mathbf{\Theta}}] &= \hat{\mathbf{C}}(\hat{\mathbf{B}}^2\hat{\mathbf{\Sigma}}_t + \hat{\sigma}_h^2\mathbf{I})\hat{\mathbf{C}}^T + \hat{\sigma}_f^2\mathbf{I} \\ &- \hat{\mathbf{C}}\hat{\mathbf{B}}\hat{\mathbf{\Sigma}}_t\hat{\mathbf{W}}^T(\hat{\mathbf{W}}\hat{\mathbf{\Sigma}}_t\hat{\mathbf{W}}^T + \hat{\sigma}_e^2\mathbf{I})^{-1}\hat{\mathbf{W}}\hat{\mathbf{\Sigma}}_t\hat{\mathbf{B}}\hat{\mathbf{C}}^T, \end{aligned} \tag{28}$$

where $\hat{\mathbf{\Theta}} = (\hat{\mathbf{W}}, \hat{\mathbf{C}}, \hat{\mathbf{B}}, \hat{\mathbf{\Sigma}}_t, \hat{\sigma}_e^2, \hat{\sigma}_f^2, \hat{\sigma}_h^2)$ are the estimated model parameters.

To quantify the uncertainty in the predictions, we can compute the credible intervals using the mean and covariance matrix of the predictive distribution using by Algorithm. 3.

Algorithm 3 Credible interval computation

Estimate the model parameters $\hat{\mathbf{\Theta}} = (\hat{\mathbf{W}}, \hat{\mathbf{C}}, \hat{\mathbf{B}}, \hat{\mathbf{\Sigma}}_t, \hat{\sigma}_e^2, \hat{\sigma}_f^2, \hat{\sigma}_h^2)$ using the SLM algorithm.
for each new observation data point $\mathbf{x}_{new}$ **do**
  Compute the mean $\mathbb{E}[\mathbf{y}_{new}|\mathbf{x}_{new}, \hat{\mathbf{\Theta}}]$.
  Compute the covariance matrix $\mathrm{Cov}[\mathbf{y}_{new}|\mathbf{x}_{new}, \hat{\mathbf{\Theta}}]$.
  for each element j of $\mathbf{y}_{new}$ **do**
    Compute the lower bound of the credible interval: $\mathrm{CI}_j^{\text{lower}} = \mathbb{E}[\mathbf{y}_{new,j}|\mathbf{x}_{new}, \hat{\mathbf{\Theta}}] - z_{\alpha/2}\sqrt{\mathrm{diag}(\mathrm{Cov}[\mathbf{y}_{new}|\mathbf{x}_{new}, \hat{\mathbf{\Theta}}])_j}$
    Compute the upper bound of the credible interval: $\mathrm{CI}_j^{\text{upper}} = \mathbb{E}[\mathbf{y}_{new,j}|\mathbf{x}_{new}, \hat{\mathbf{\Theta}}] + z_{1-\alpha/2}\sqrt{\mathrm{diag}(\mathrm{Cov}[\mathbf{y}_{new}|\mathbf{x}_{new}, \hat{\mathbf{\Theta}}])_j}$
  end for
end for

Table 5. Percentage of elements within prediction interval for different alpha values and folds. All the data are presented as %.

Alpha	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
0.05	95.15	95.05	94.25	93.60	95.65
0.10	90.08	89.70	89.38	88.33	91.05
0.15	84.98	84.65	84.10	83.25	85.83
0.20	80.35	79.85	79.23	78.08	80.73
0.25	75.15	74.88	74.30	73.15	75.68

To validate the effectiveness of our PPLS prediction method, we conduct a comprehensive simulation experiment. We generate synthetic data using the PPLS model structure, with 200 features each for the observation data X and target data Y, 50 latent variables, and 100 samples. The experiment employs a 5-fold cross-validation approach, with the true model parameters to make predictions on the test set, simulating an ideal scenario where the model is perfectly calibrated. We compute credible intervals for the predictions using various alpha values (0.05, 0.10, 0.15, 0.20, and 0.25) to assess the method's performance across different confidence levels.

The results (Table 5) demonstrate the high accuracy and consistency of our proposed PPLS model combined with the SLM algorithm. For each alpha value, remarkably consistent performance across all five folds demonstrates the stability of our approach. As the alpha value increases, the percentage of elements within the prediction interval decreases, which corresponds to the narrowing of the credible interval. Most notably, the observed percentages closely align with the expected confidence levels (e.g., 95% for alpha=0.05, 90% for alpha=0.10, etc.). This alignment indicates that our method yields well-calibrated uncertainty estimates – a critical feature for reliable predictions in real-world applications, provided the underlying data structure adheres to the PPLS model assumptions.

8 Conclusion

By utilizing the scalar likelihood function and a rank n update technique, we propose a Scalar Likelihood Method (SLM), which provides an efficient and accurate alternative to the traditional EM algorithm. The theoretical contributions of our work include a closed-form expression of the log-likelihood function in terms of scalar variables, as well as closed-form estimators for the noise distributions. The proposed algorithm enhances the interpretability and reliability of the model. Empirical applications in association analysis and predictive modeling demonstrate SLM's superior performance and its potential for uncovering new insights in fields such as genomics. In the future, we plan to explore more general forms of the PPLS model.

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